

**Topic:** Cross-modal hallucination in vision-language models with degraded image inputs

**Description:** We are basically trying to judge how degraded images increase hallucinations in vision-language models.

Vision-language AI models generate false textual descriptions when image quality is poor, so we can measure the severity of the hallucinations with different levels of degradation and if presence of text in prompt increases this hallucination.

### **The gap this study fills -**

No published study has (a) systematically applied a degradation pipeline to VLM inputs, (b) measured hallucination rates as a function of degradation type and severity, (c) tested whether textual context amplifies hallucination under degradation, or (d) produced a qualitative taxonomy of failure modes under visual degradation. This proposal addresses all four gaps in a single unified study.

### **Research Questions:**

1. RQ1 (Prevalence): How do hallucination rates — measured by object hallucination (CHAIR), factual error rate, and hallucination severity — change as image quality degrades across seven degradation categories?
2. RQ2 (Model variation): Which VLMs are most and least robust to visual degradation? Do architectural choices (encoder type, training data scale, RLHF alignment) predict degradation robustness?
3. RQ3 (Context amplification): Does the presence of a descriptive text prompt — providing contextual cues about the image content — increase hallucination rates under degraded conditions, as the model substitutes textual priors for missing visual signal?
4. RQ4 (Failure taxonomy): Can hallucinations under degradation be classified into a meaningful taxonomy — e.g., object confabulation, attribute substitution, spatial relation errors, count errors — and do different degradation types produce distinct failure signatures?

### **Target Models :**

GPT-4V, Gemini 1.5 Pro, LLaVA-1.6, InternVL, LLaMA3

### **Primary Tools:**

Python, OpenCV, CHAIR metric, GPT-4 judge, POPE benchmark

## **Degradation Type:**

Gaussian blur  
JPEG compression  
Gaussian noise  
Partial occlusion  
Resolution downscale  
Fog/haze simulation  
Salt-and-pepper noise

## **Datasets:**

1. MSCOCO
2. NoCaps
3. HallusionBench